

UNIT II – OVERVIEW OF DATA MINING ALGORITHMS

1. Definition of Models and Patterns

- A **model** is a mathematical or logical representation of data.
- It captures relationships between input and output variables.

Examples:

- Decision Tree
- Regression Equation
- Neural Network
- A **pattern** is a meaningful and interesting relationship discovered in data.

Examples:

- “Students with attendance > 80% usually pass”
- “Customers who buy X also buy Y”

□ Key Point:

- Model → structure
 - Pattern → knowledge extracted
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2. Reductionist Approach

- Real-world data is complex and high-dimensional.
- Reductionist approach simplifies data by:
 - Removing irrelevant features
 - Selecting important attributes
 - Reducing noise

Techniques:

- Feature Selection
- Dimensionality Reduction (PCA)
- Sampling

□ **Advantage:**

- Faster computation
 - Better model accuracy
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3. Score Functions

- Score functions measure **how good a model is**

Common Score Functions:

- Accuracy
- Error Rate
- Likelihood
- Information Gain

Example:

- Model A Accuracy = 85%
 - Model B Accuracy = 92%
 - Model B is better
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4. Types of Models

A. Predictive Models

- Used to predict unknown values

Examples:

- Classification → Pass/Fail
 - Regression → Marks prediction
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B. Descriptive Models

- Used to find hidden patterns

Examples:

- Clustering → Student groups
 - Association → Market basket analysis
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5. Model Structures

A. Regression

- Predicts continuous values
 - Example: Marks prediction
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B. Classification

- Predicts categories
 - Example: Pass / Fail
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C. Clustering

- Groups similar data
 - Example: Student performance groups
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6. Probability Models and Density Estimation

- Based on probability theory

Examples:

- Naive Bayes
- Gaussian Distribution

Density Estimation:

- Estimates data distribution

□ Used in:

- Spam filtering
- Risk prediction

7. Curse of Dimensionality

- Occurs when number of features increases

Problems:

- Data becomes sparse
- Distance measures become unreliable
- Model accuracy decreases

Example:

- 2D → easy
- 100D → complex

Solution:

- Feature selection
 - Dimensionality reduction
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8. Models for Structured Data

A. Sequence Data

- Time-based data
- Example: Student attendance over time

B. Graph Data

- Relationships between entities
 - Example: Social networks
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9. Evaluation Metrics

Accuracy

- Correct predictions / Total

Precision

- Correct positive predictions

Recall

- Ability to find all positives

F1 Score

- Balance of precision & recall
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10. Robust Methods

- Handle real-world issues:

A. Noise Handling

- Remove incorrect data

B. Missing Values

- Fill using:
 - Mean
 - Median
 - Prediction

C. Outliers

- Detect and remove anomalies
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Conclusion

- Data mining algorithms rely on:
 - Models
 - Evaluation metrics
 - Efficient processing
- Proper model selection leads to:
 - Better prediction
 - Meaningful insights

